深圳市拓淘电子科技有限公司
Shenzhen Toppop Electronic Co.,Ltd

TOPPOP

LCM APPROVAL SHEET
模组承认书

版本(Version): V0

Project No. 项目型号	TT191RGN10A		
Customer 客户名称			
Module No. 客户型号			
Product type 产品类型	Type : AMOLED Display Resolution : 240x536 Dots Screen Size : 1.91 inch		
Signature by customer: 客户确认签字盖章：			
Structure size: 结构尺寸：	☐ OK_____ ☐ NG_____	备注：	
Electric property: 电气性能：	☐ OK_____ ☐ NG_____	备注：	

Company 公司	Designed by 设计	Checked by 审核	Approved by 批准
Signature 签名			

Rev. 版本	Date 日期	Description 变更内容
V0	2017-10-25	Preliminary Specification Release

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1. Scope

2. DEFINITION OF TERM

Min.

Case of output value : The minimum value to occur by a mass production
Case of input value : The minimum vale to satisfy the specification
Case of mechanical value : The minimum value to occur by a mass production
Min value is guaranteed value.

Typ.

Central value
Typ. Value is not guaranteed value.

Center.

Average value of mass production.

Max.

Case of output value : The maximum value to occur by a mass production
Case of input value : The maximum vale to satisfy the specification
Case of mechanical value : The maximum value to occur by a mass production

Br000 to Br255 (Only AMOLED)

Adjustment level of the brightness.
B000 is min brightness, B255 is max brightness.

V000 to V255

Adjustment level of the White color gray scale.
V000 is min brightness, V255 is max brightness.

R000 to R255

Adjustment level of the Red color gray scale.
G000 is min brightness, G255 is max brightness.

G000 to G255

Adjustment level of the Green color gray scale.
G000 is min brightness, G255 is max brightness.

B000 to B255

Adjustment level of the Blue color gray scale.
B000 is min brightness, B255 is max brightness.

Worst case pattern

The test pattern that specification have worst.

Worst case condition

The test condition that specification have worst.

ALL angle

Omnidirectional angle
θ = 0deg. to 88deg.
Φ = 0:00(12:00) to 12:00. *360degree.

Reference and limit sample list

Supplier necessary follow this reference and limit sample.

3. Features

3.1 Product Applications Smart Phone

3.2 Product Features

1)Display color: 16.7M (RGB x 8bits)
2)Display format: 1.91”(240RGBx536)
3)Pixel arrangement: Real RGB arrangement
4)Interface: MIPI(1Lane)

3.3 Model Name: TT191RAN10A

Code	Definition	Description
191	Display Size	1.91inch
8	Resolution	240RGB x 536
A	Display Panel Key Technology	LTPS, Real RGB,Thinning
M	Product Status	FOG（Not Oncell）
3	Customer Code	

4. Mechanical Specifications

Item	Specification	unit
LTPS Glass outline	22.40 x 51.32	mm
Number of dots	240(W)x RGB x 536(H)	dots
Active area	19.80x44.22	mm
Diagonal size	1.91	inch
Pixel pitch	82.5* 82.5	μm
Glass thickness（LTPS/Encap. glass）	0.3 / 0.2	mm
Weight	2.16	g（TYP）

5. Maximum Rating

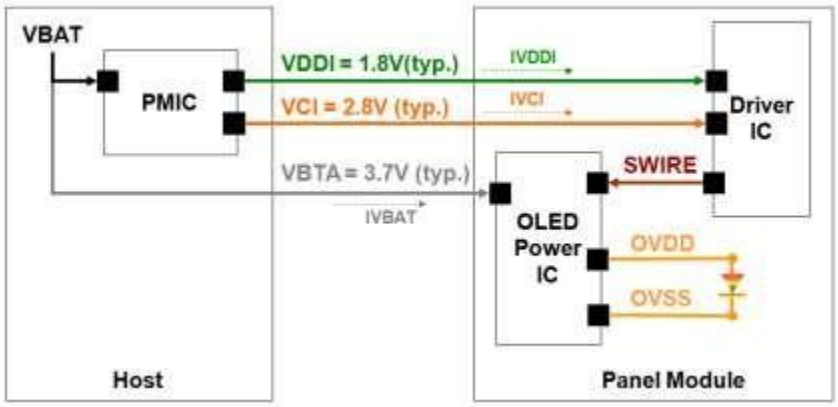
Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit	Remark
Battery supply	VBAT		-0.3	-	6	V	
Power supply for Logic	VDDIO		-0.3	-	5.5	V	
Power supply for Analog	VCI		-0.3	-	5.5	V	

6. Electrical Specifications

6.1 Electrical Characteristics

6.1.1 Power Characteristic：

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit	Remark
Battery supply	VBAT		2.5	3.7	4.5	V	
Power supply for Logic	VDDIO		1.65	1.8	1.95	V	
Power supply for Analog	VCI		2.65	2.8	3.6	V	



1) Normal Mode

Power Supply: VDDIO=1.8V VCI=2.8V Vbat=3.7V

Frame Frequency: Fframe =60HZ @ 25degC, Brightness 350 nits, Command Mode,

Parameter	Symbol	Conditions	Min.	Typ.	Max	Unit	Remark
Current for OLED	Ivbat		-	42	50.4	mA	
Current for VDDIO	Ivddio		-	6.5	7.8	mA	
Current for VCI	Ivci		-	12	14	mA	

2) Idle Mode

Power Supply: IOVCC=1.8V VCI=2.8V

Frame Frequency: Fframe =15HZ @ 25degC, Brightness 50 nits,

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
10% Pixel On 50 nits	Ivbat	-	-	-	mA	Supplied by Driver IC
	IVCI	-	6	7.2	mA	Ref
	IVDDIO	-	1	1.2	mA	Ref

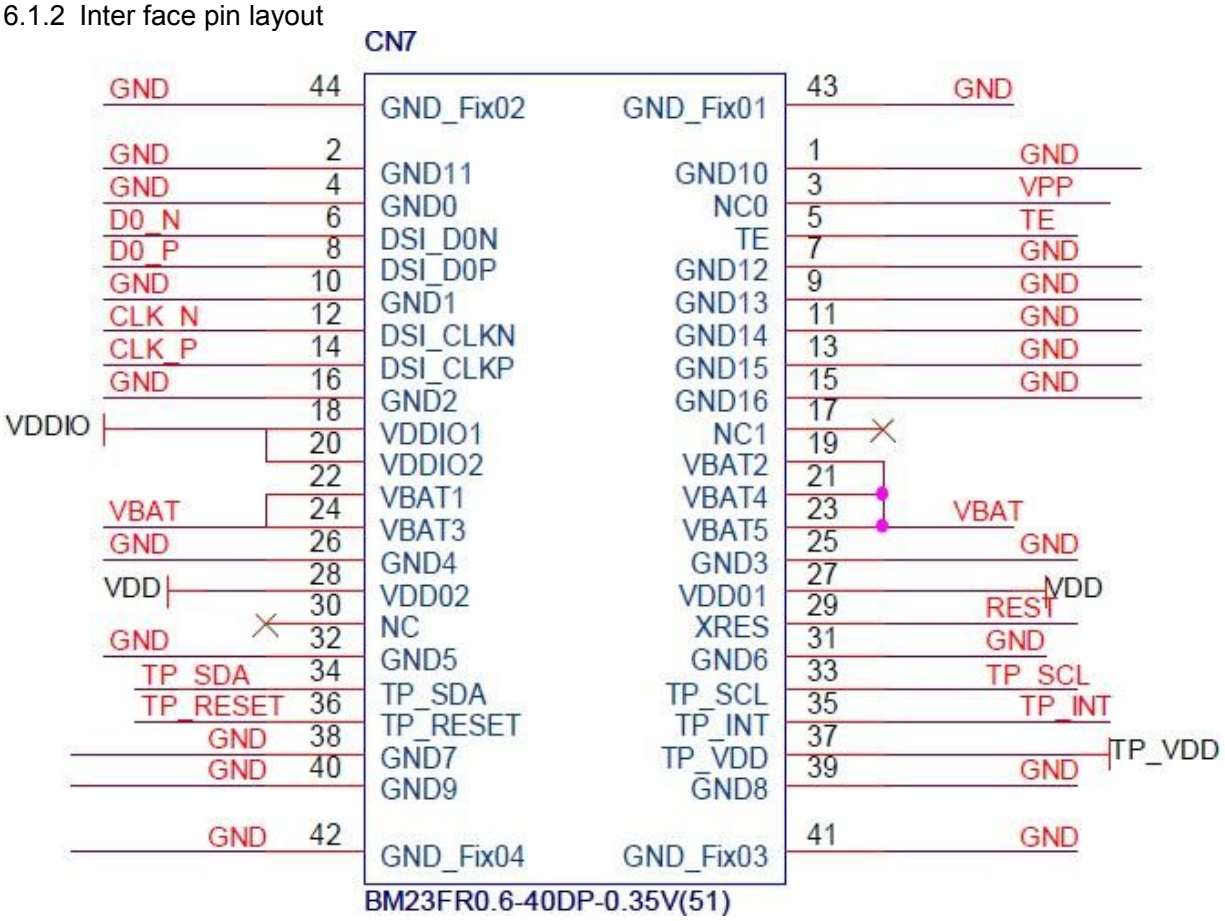
3) Sleep IN Mode

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
Sleep IN Mode	IVCI	-	0.25	0.3	mA	-
	IVDDIO	-	0.1	0.12	mA	-

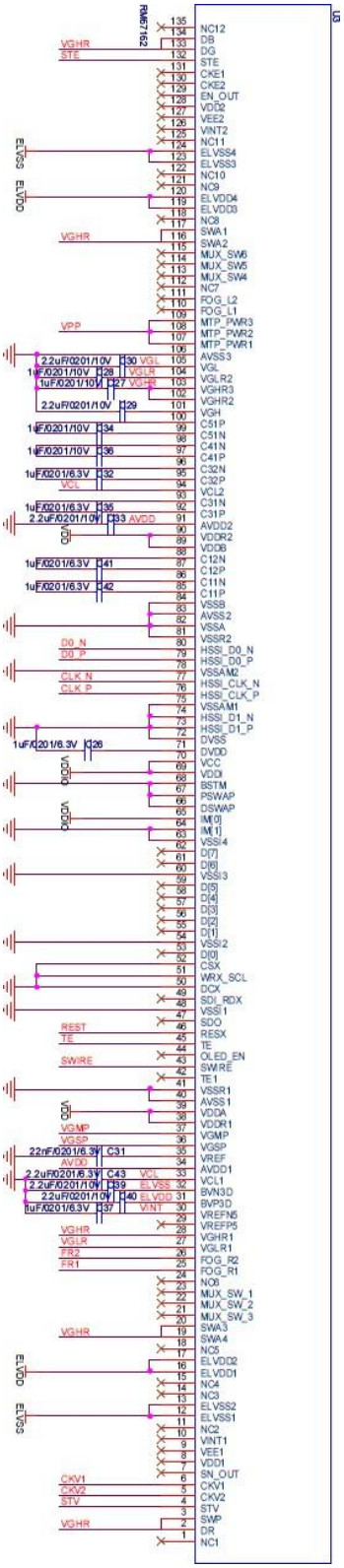
4) Deep Standby Mode

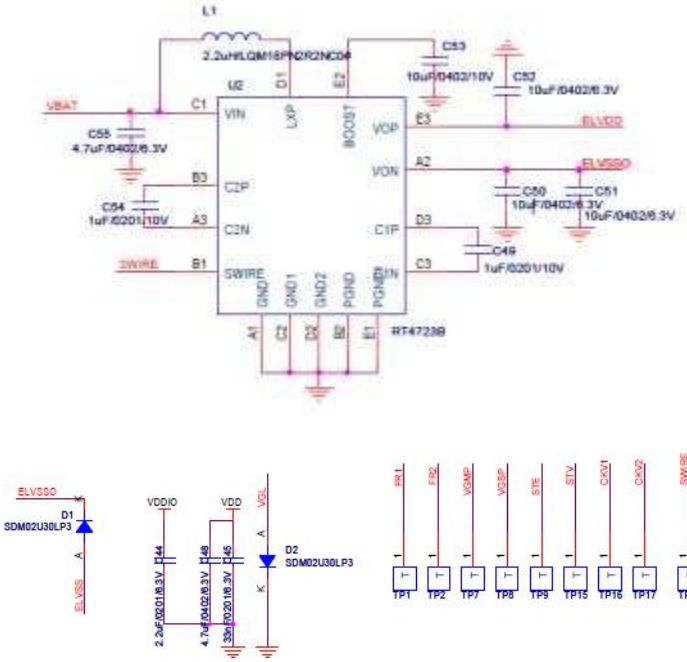
Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
Deep Standby	IVCI	-	-	2	uA	-
	IVDDIO	-	-	1	uA	-

6.1.2 Inter face pin layout



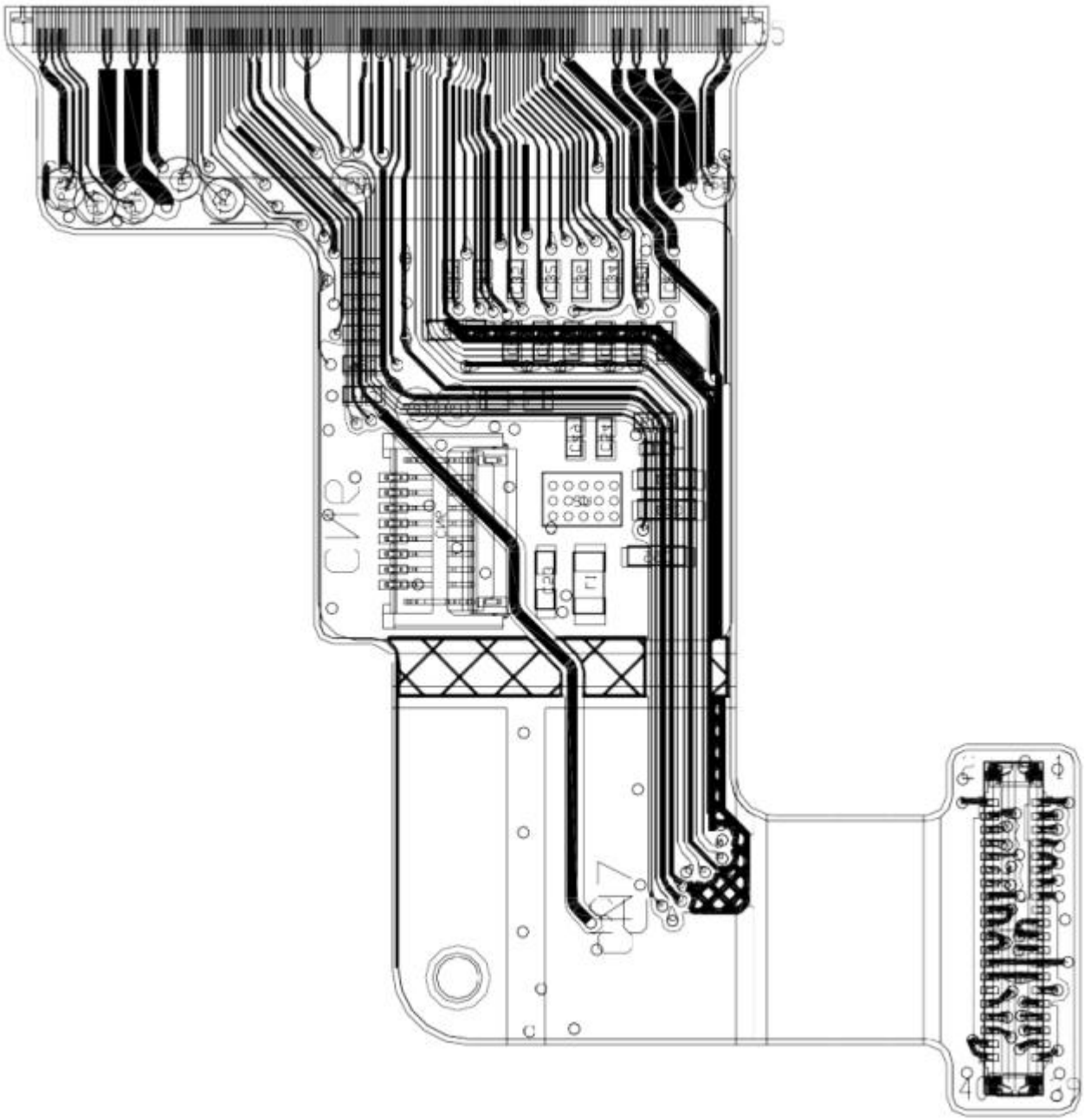
6.1.3 FPC Diagram



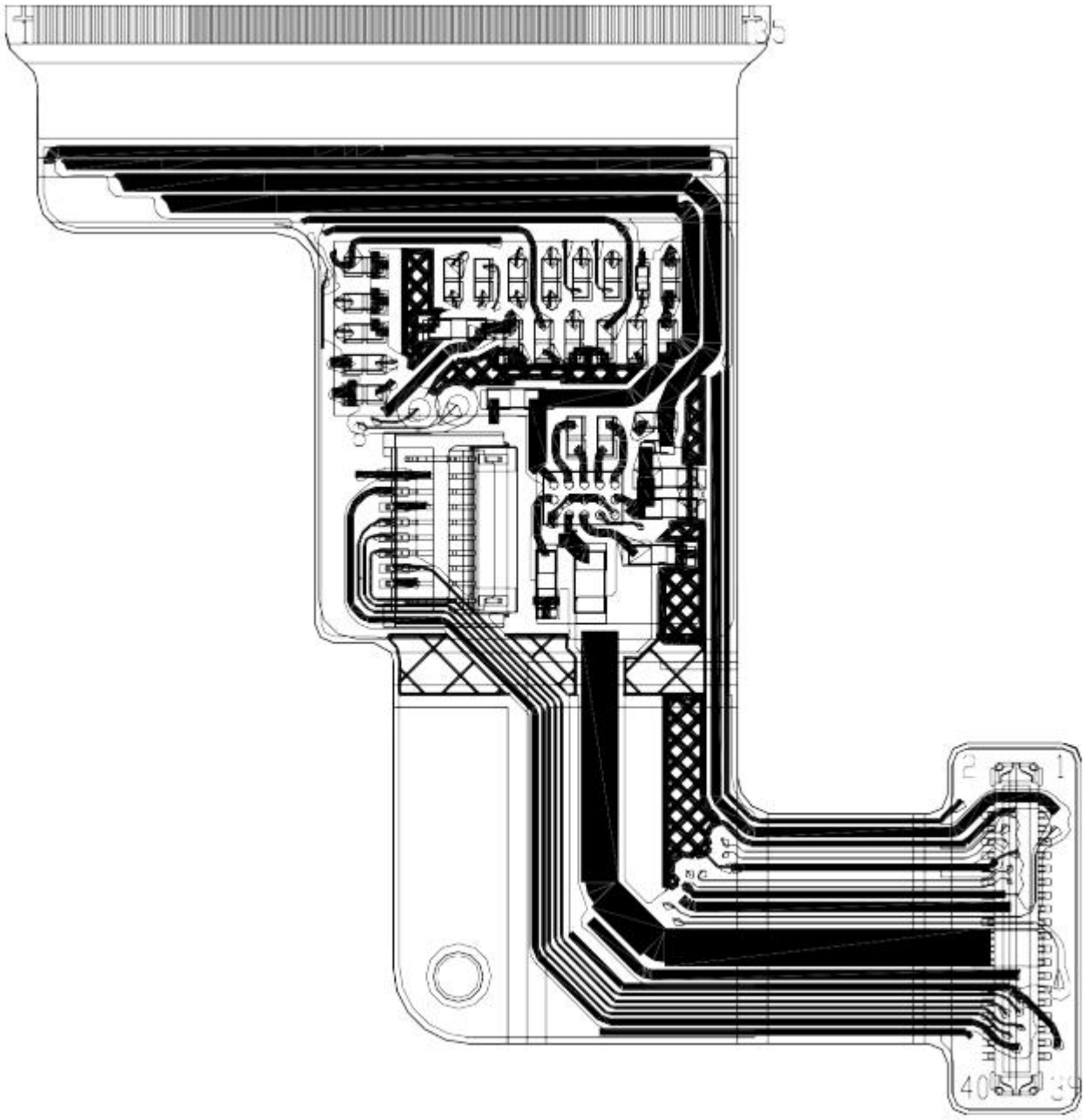


6.1.4 FPC Layout

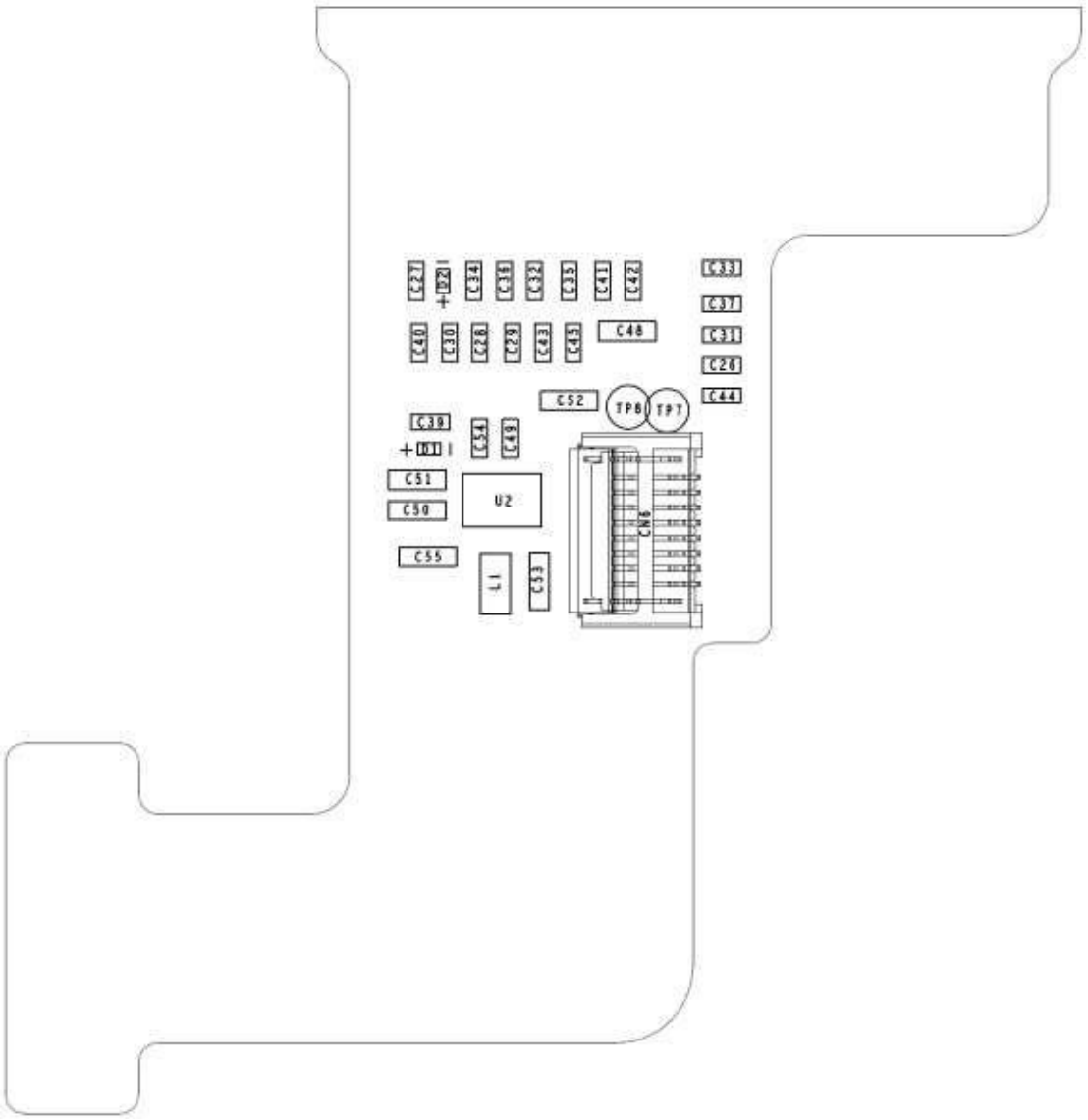
TOP



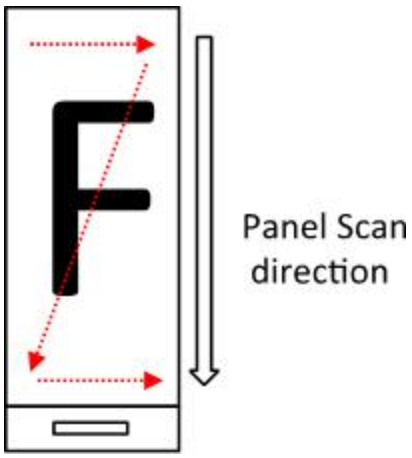
Bottom



Components Position

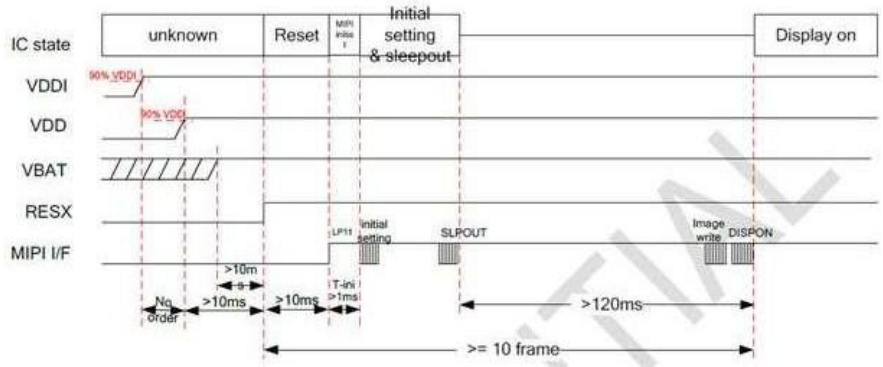


6.1.5 Graphic memory writing direction

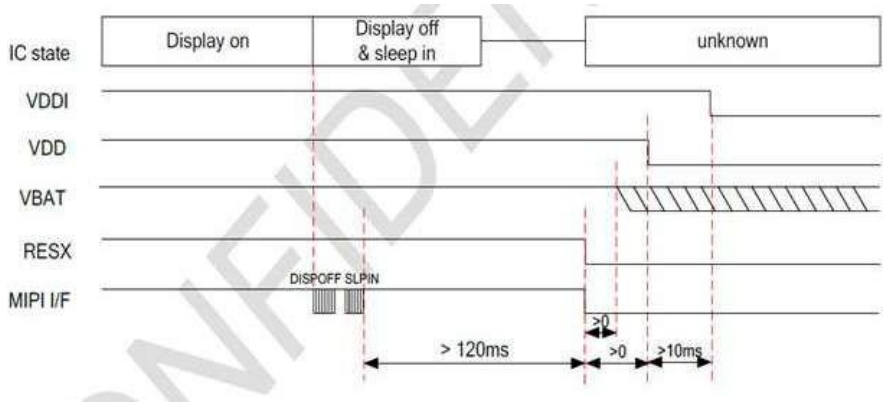


6.2 Recommended Operating Sequence

6.2.1 Power on sequence



6.2.2 Power off sequence



6.3 Initial code

RM67162-MEIZU-QVA2-V04				
W/R	Type	Register	Parameter	Description
Turn on VDDIO				
Delay (No Limit)				
Turn on VCI				
Delay (No Limit)				
Turn on VBAT				
Delay >10ms				
Reset pin high				
Delay >10ms				
MIPI初始化				
Delay >1ms				
w	0x15	0xFE	0x04	PAGE 3
w	0x15	0X6A	0X00	
w	0x15	0xFE	0x05	设定Page 4
w	0x15	0x05	0x00	ovss: -2.4(根据power调整)
W	0x15	0xFE	0x07	Page6
W	0x15	0x07	0x4F	
w	0x15	0xFE	0x01	Page 0
W	0x15	0x2A	0x02	
w	0x15	0x2B	0x73	
w	0x15	0xFE	0x0A	Page9
w	0x15	0x29	0x10	
w	0x15	0xFE	0x00	设定user command
w	0x15	0x35	0x00	
W	**0x15**	**0x51**	**0x00**	**2017/4/25在V04版上新增此行代码 (Only for 客户平台端),即先设定亮度到最低**
w	0x15	0x11	sleep out	
Delay >120ms				
w	0x15	0x29	display on	
HFP	20	VFP	20	
HBP	40	VBP	12	
HPW	20	VPW	4	
HPX	240	VPX	536	
HAA	HBP+HPX	VAA	VBP+VPX	
HAL	HFP+HBP+HPX	VAL	VFP+VBP+VPX	

2. Power Off Sequence

W/R	Type	Register	Parameter	Description
w	0x15	0x28	0x00	Display Off
w	0x15	0x10	0x00	Sleep in
Delay >120ms				
Reset pin low				
Dealy >0ms				
Turn off VBAT				
Delay >10ms				
Turn off VC& VDDIO				

3. Sleep In

W/R	Type	Register	Parameter	Description
w	0x15	0x28	0x00	Display Off
w	0x15	0x10	0x00	Sleep in
Delay >120ms				

4. Sleep Out

W/R	Type	Register	Parameter	Description
w	0x15	0x11	0x00	Sleep Out
Delay >120ms				
w	0x15	0x29	0x00	Display On

5. Idle mode In

W/R	Type	Register	Parameter	Description
w	0x15	0xFE	0x00	
w	0x15	0x39	0x00	Idle In

6. Idle mode Out

W/R	Type	Register	Parameter	Description
w	0x15	0xFE	0x00	
w	0x15	0x38	0x00	Idle Out

7. Deep standby In

W/R	Type	Register	Parameter	Description
w	0x15	0x4F	0x01	Pull CLK_P/N & D0_P/N to GND

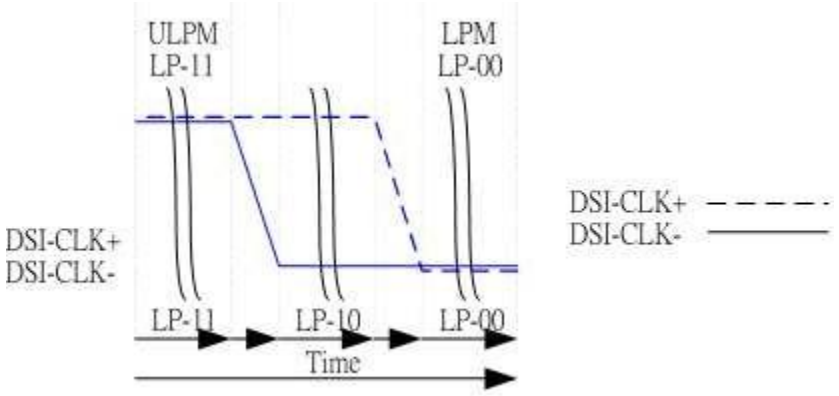
8. Deep standby Out

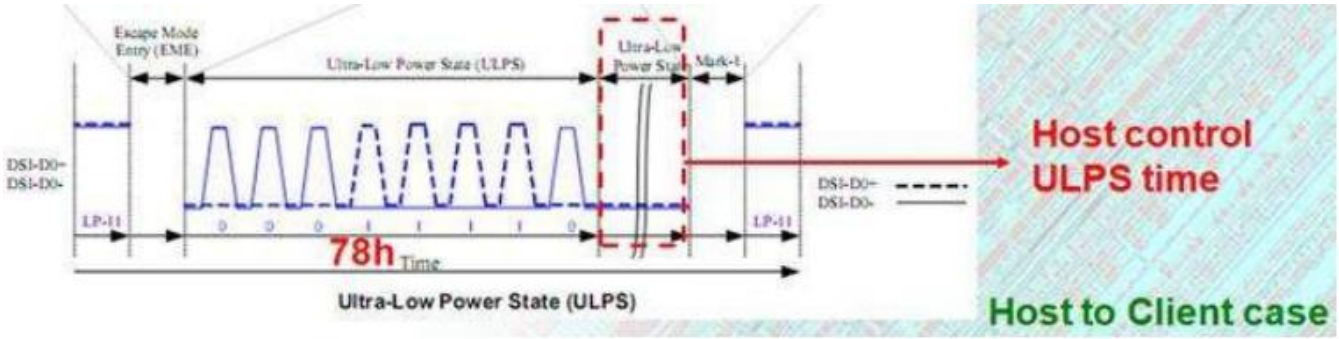
W/R	Type	Register	Parameter	Description
				Rset pin low
Delay >3ms				
				Rset pin high
				Power on Sequence

Note1:
There is no need to enter sleep in before entering deep Standby. After 4F 01, MIPI CLK and Data can be lowered without delay requirement. After entering deep Standby, Reset is still high. VDDIO,VCI, and Vbat do not need to be disconnected.



Note2: Enter ULPS





6.4 ID code request

	D7	D6	D5	D4	D3	D2	D1	D0	Remark
ID1	P17	P16	P15	P14	P13	P12	P11	P10	Read DAhRegister
ID2	P27	P26	P25	P24	P23	P22	P21	P20	Read DBh Register
ID3	P37	P36	P35	P34	P33	P32	P31	P30	Read DCh Register

ID1: Production code

P17-P16 Manufacture code
00:
01
P15-P10 Production date
000000: Sample stage
000001: MP 1st week
000010: MP 2nd week
111111: MP 63th week

Serial number should be relate to ID1.

Note：a. MP The first week of mass production was coded as 01 weeks

- If there is no production in the next week (the code corresponds to week 02), the corresponding week code will be invalid
- If there is production in the next week (code corresponding to 03 weeks), the corresponding week code will be burned to 03 weeks

ID2: Revision code
P27-P26 Factory code
Supplier proposal
P25 Sample stage
0: sample
1: MP
P24-20 Revision code

ID3: device information code
P37-P30 Production date
0000 1111：Line5(G2.5)
0000 1101：Line2(G3.5)

7 Electro-Optical Specification

Optical characteristics

Item		Symbol	Condition	Min	Typ	Max	Unit	Remark
Contrast Ratio		CR0	θ=0deg.	10000	-	-	-	
Brightness (φ 12:00,3:00,6:00,9:00)		Br0	Θ=0deg.	315	350	385	cd/m2	
		Br45	Θ=45deg.	115	150	-		
NTSC Ratio		S0	Θ=0deg.	95	105	-	%	ICE1931
Adobe cover Ratio		S0R	ICE1931	92	100	-	%	Note 1
Color Temp		T		7000	7800	8600	K	
White chromaticity	X	-	θ=0deg	0.275	0.295	0.315	-	
	Y	-	θ=0deg	0.295	0.315	0.335		
Red chromaticity	X	-	θ=0deg	0.635	0.665	0.695		
	Y	-	θ=0deg	0.304	0.334	0.364		

Measurement method.

Measurements condition as below, if not otherwise specified. Include touch panel, OCA and Cover glass

Room temp: 25°C, Frame frequency=60Hz Image Enhancement :OFF

Measurement points: Display center, Θ=0 deg.

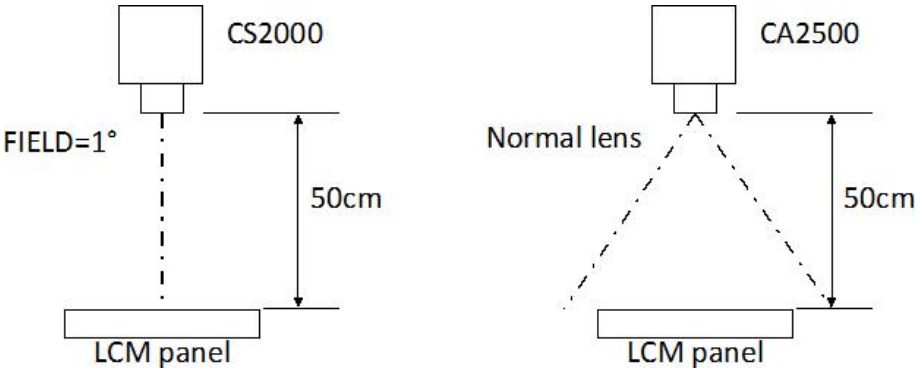
Measurement instrument: Uniformity & Crosstalk CS2000, Flicker&Gamma CA310.

Other items CS2000 or CA310。

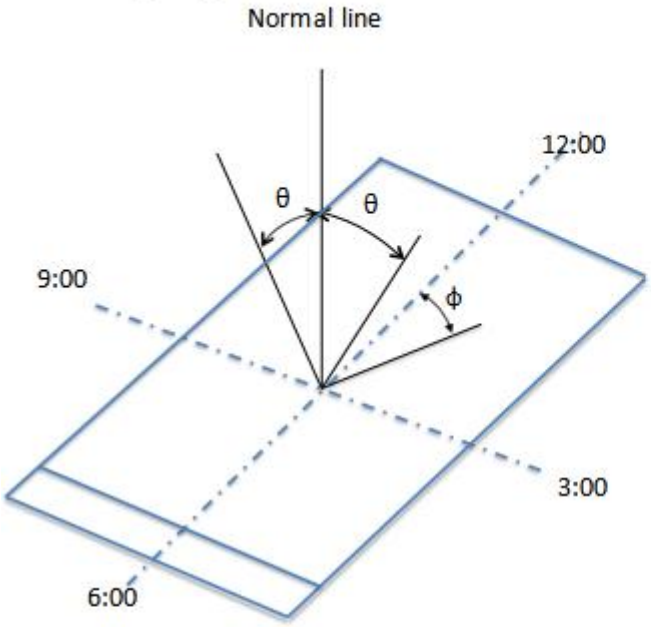
CS2000:To be measured on the center area of Panel with a viewing cone of 1 ° by luminance mater, after 15min operation

CA2500:To be measured on the Active area of Panel with a viewing cone of 35pixel/circle by luminance mater, after 15min operation

CA310: To be measured on "CA-P32/35" Probe



[Viewing angle]



Note1) Define of Adobe cover ratio

Green: RGB color chromaticity of this module

Rad : RGB color chromaticity of Adobe RGB

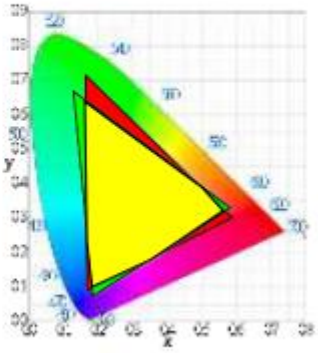
R: x0.64, y0.330

G: x0.21, y0.71

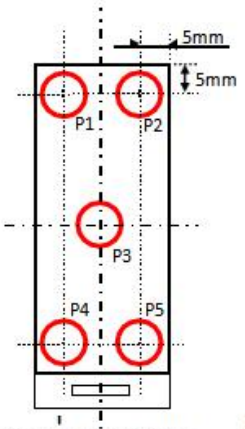
B: x0.15, y0.06

Yellow : The area where red and yellow are piled

Adobe RGB cover Ratio = Yellow / Rad *100[%]



Note2) Define of Brightness uniformity and Color uniformity



$$\text{Brightness uniformity} = \frac{\text{Minimum value[P1:P5]}}{\text{Maximum value[P1:P5]}}$$

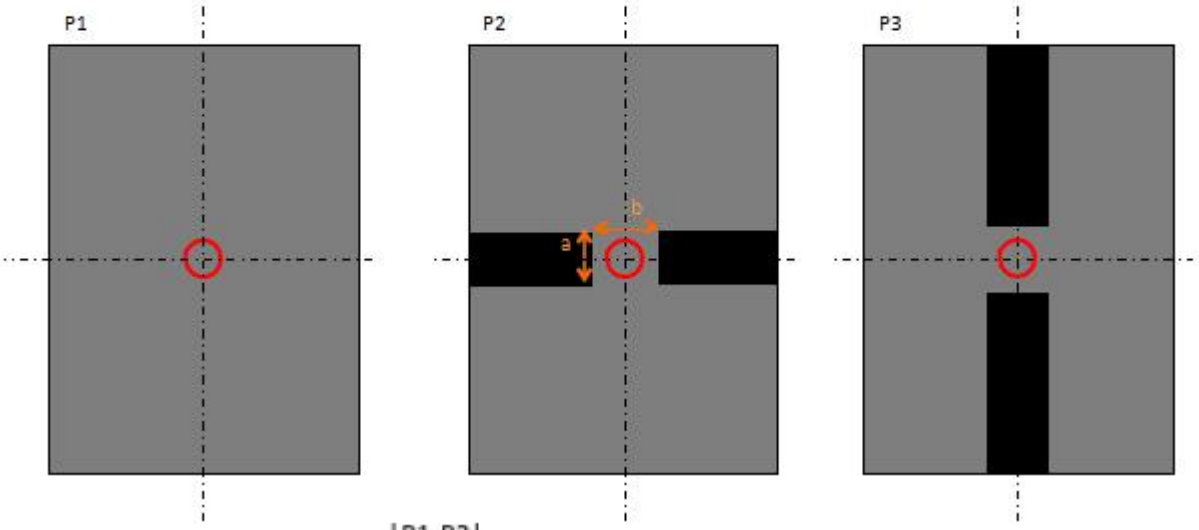
Color uniformity = Maximum value[P1:P5] - Minimum value[P1:P5]

Note3) Define of crosstalk

Base color : V127

Measurement area (a,b) : 144dots*144dots

Bar color : white, red, green, blue, Black.



$$\text{Cross talk ratio P2} = \frac{|P1-P2|}{P1}$$

$$\text{Cross talk ratio P3} = \frac{|P1-P3|}{P1}$$

Cross talk ratio : maximum value of cross talk P2 and P3

Note4) Define of White color shift

White color shift is Maximum value of Color shift WADu' and Color shift WADv'

$WADu' = |u'_{0} - u'_{45}|$

$WADv' = |v'_{0} - v'_{45}|$

$\triangle u'v' = \sqrt{WADu'^2 + WADv'^2}$

u'_{0}, v'_{0} : white color chromaticity at $\Theta = 0deg$

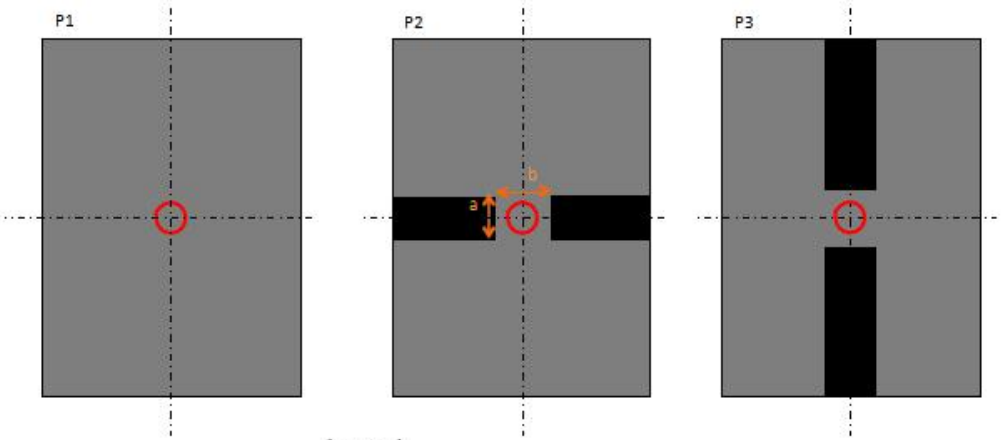
u'_{45}, v'_{45} : white color chromaticity at $\Theta = 45deg$ *(φ=all angle)

Note3) Define of crosstalk

Base color : V127

Measurement area (a,b) : 144dots*144dots

Bar color : white, red, green, blue, Black.



$$\text{Cross talk ratio P2} = \frac{|P1-P2|}{P1}$$

$$\text{Cross talk ratio P3} = \frac{|P1-P3|}{P1}$$

Cross talk ratio : maximum value of cross talk P2 and P3

Note4) Define of White color shift

White color shift is Maximum value of Color shift WADu' and Color shift WADv'

WADu' =|u' 0 - u' 45|

WADv' =|v' 0 - v' 45|

$\triangle u'v' = \sqrt{WADu'^2 + WADv'^2}$

u' 0,v' 0 : white color chromaticity at Θ=0deg

u' 45,v' 45 : white color chromaticity at Θ=45deg *(φ=all angle)

8 Life Time

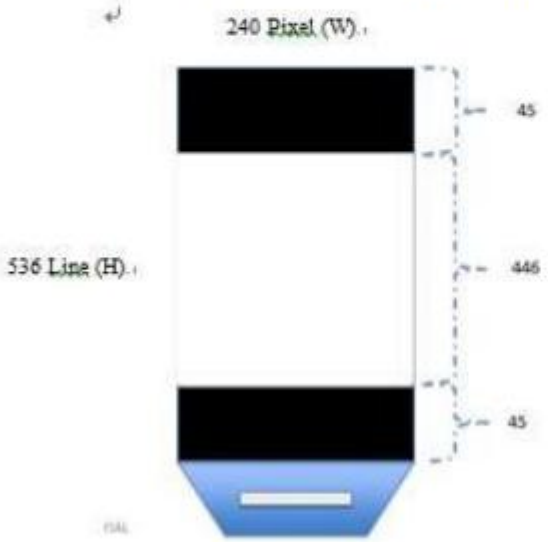
Parameter	Symbol	Conditions	Min.	Center	Max	Unit	Remark
Life time (1)		Frame rate =60Hz	93	-	-	%	Note 1
Life time (2)		Frame rate =15Hz Brightness 50nit	98			%	
		Frame rate =30Hz Brightness 50nit					
Image sticking time		Frame rate =60Hz Brightness 350nit	24	-	-	h	Note 2
Image sticking time		Frame rate =30 or 15Hz Brightness 50nit	48			h	

Note1. Define Life time

The brightness after 240h

Room Temp : 25°C
Test pattern : all white
Life time = Center Brightness of after 240h / Center Brightness initial value *100[%]

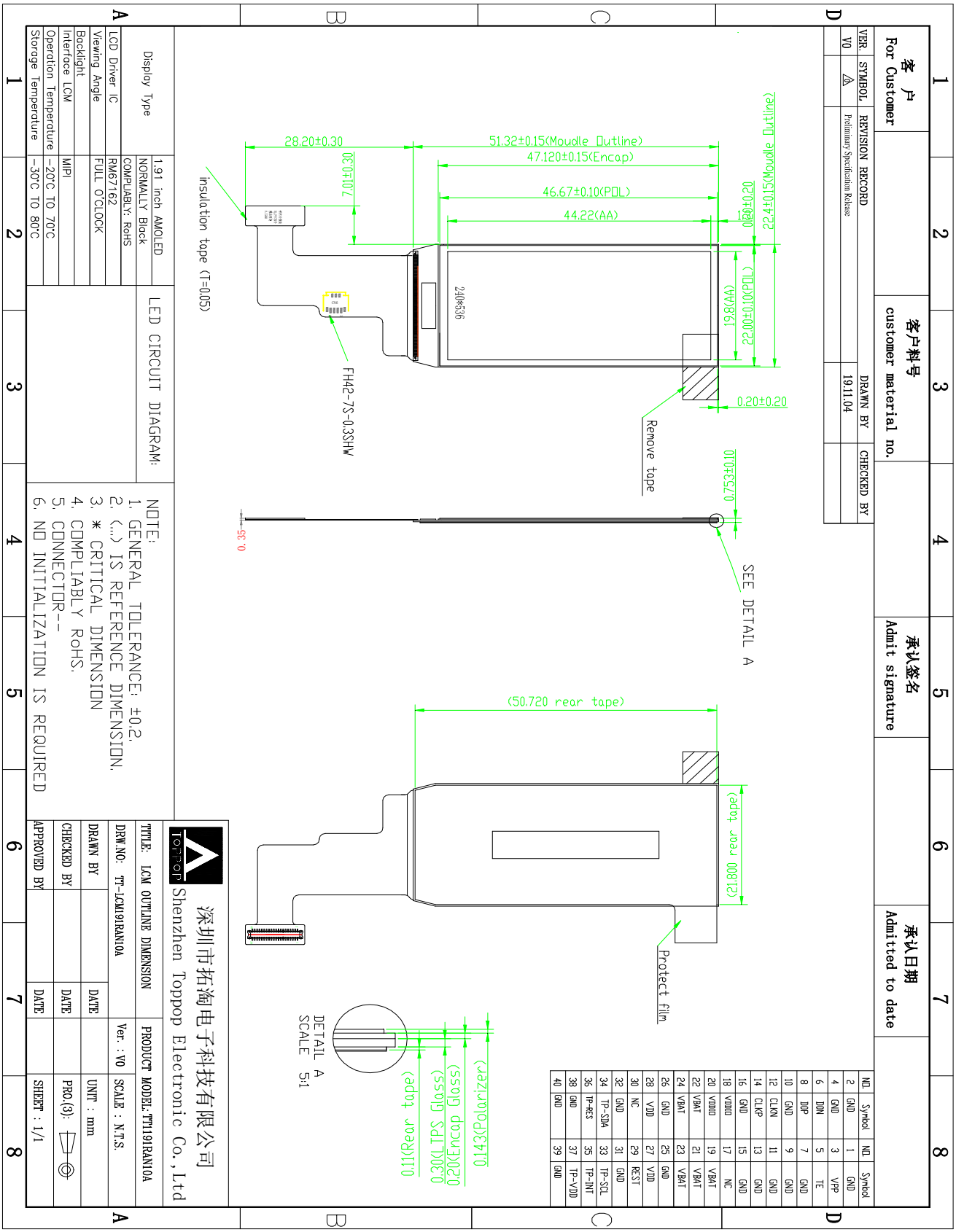
Note2. Define image sticking time
Time before 1% of brightness down from an initial value.
Room Temp. : 45°C
Test pattern : Base color is Black
Box color is Worst case pattern



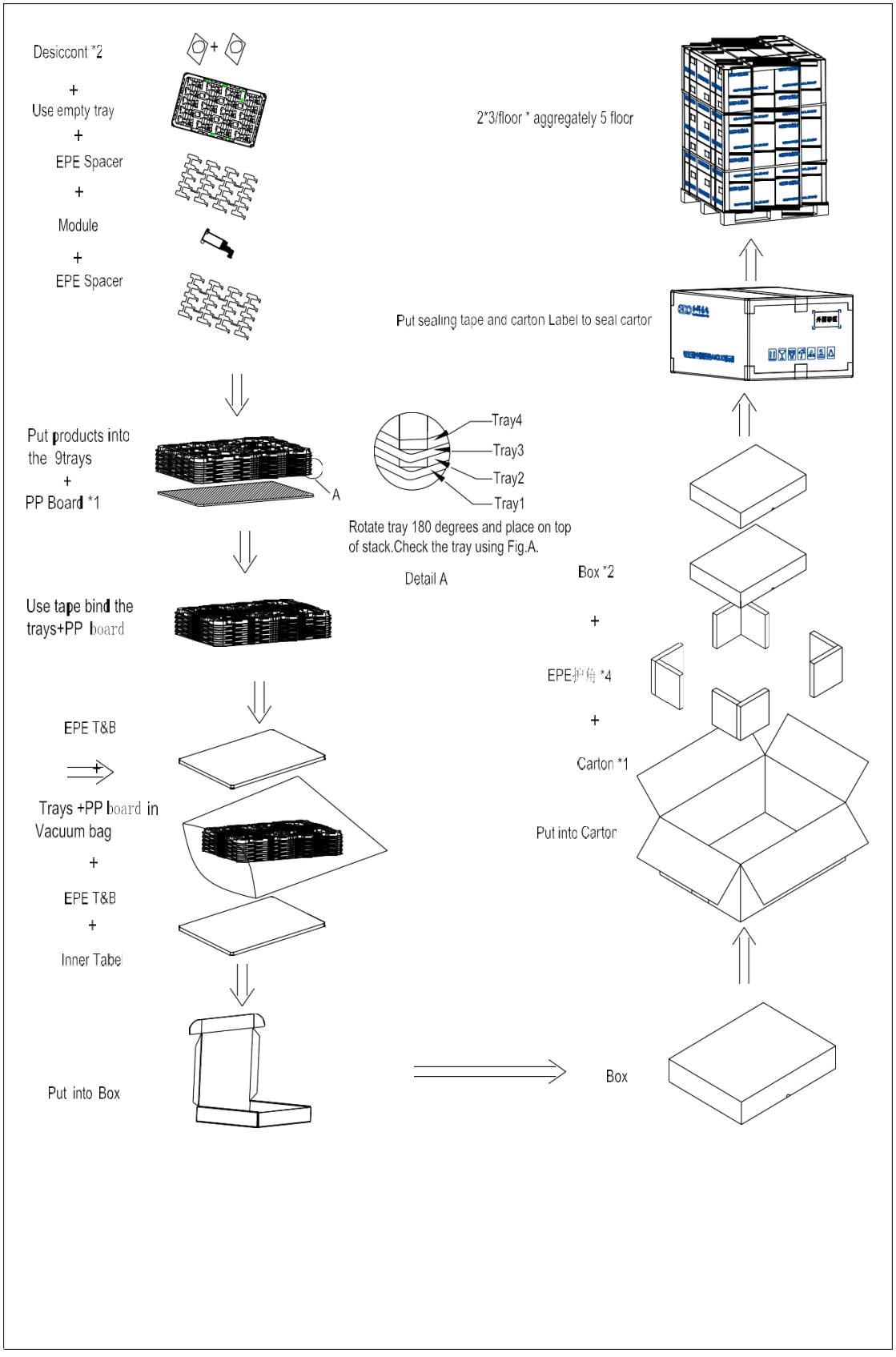
9 Reliability

-

10. Outline Dimension Drawing



11. Packing


Desiccont *2
+
Use empty tray
+
EPE Spacer
+
Module
+
EPE Spacer
Put products into the 9trays
+
PP Board *1
A
Tray4
Tray3
Tray2
Tray1
Rotate tray 180 degrees and place on top of stack.Check the tray using Fig.A.
Detail A
Use tape bind the trays+PP board
EPE T&B
+
Trays +PP board in Vacuum bag
+
EPE T&B
+
Inner Tabe
Put into Box
Box
Box *2
+
EPE护角 *4
+
Carton *1
Put into Cartor
Put sealing tape and carton Label to seal cartor
2*3/floor * aggregately 5 floor